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DEGLI STUDI
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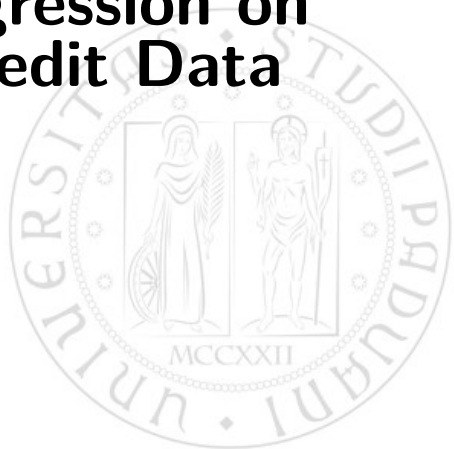
Dipartimento
di Fisica
e Astronomia
Galileo Galilei

Charm hadrons reconstruction in ALICE

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10th February 2026

Logistic Regression on German Credit Data



The Dataset

1. Original dataset contains both numerical (e.g. Age, credit amount, ...) and categorical data (e.g. Sex, Civil status, ...) → we used the numerical version provided by Strathclyde University.
2. 1000 individuals with 24 features and a classifier for "good" (+1) and "bad" (-1) creditors.

Our goal: sample from the target distribution of the posterior.

The Model

Bayesian Logistic Regression model.

The target distribution is $p(\alpha, \beta \mid X, y) \propto p(y \mid X, \alpha, \beta) p(\alpha) p(\beta)$

$$\propto \exp \left(- \sum_i \log(1 + \exp\{-y_i (\alpha + x_i \cdot \beta)\}) - \frac{\alpha^2}{2\sigma^2} - \frac{1}{2\sigma^2} \beta \cdot \beta \right)$$

where:

- x_i is a 24-dimensional vector of numerical predictors associated with a customer i .
- y_i is +1 if the customer should receive credit, -1 otherwise.
- α is a scalar intercept term.
- β is a vector of 24 regression coefficients.
- All predictors are normalized to have zero mean and unit variance.
- α and each element of β are given given weak zero-mean normal priors with a **fixed** variance $\sigma^2 = 100$.

The Model: Likelihood

We assume that the probability of a customer being "good" depends on a linear combination of the features:

$$\eta_i = \alpha + x_i \beta$$
$$p(y_i = 1 \mid x_i, \alpha, \beta) = \sigma(\eta_i)$$

where $\sigma(\eta_i) = \frac{1}{1+e^{-\eta_i}}$ is the sigmoid logistic function.

Therefore, because $y = \{-1, +1\}$

$$\begin{cases} p(y_i = 1 \mid x_i) = \sigma(\eta_i) \\ p(y_i = -1 \mid x_i) = 1 - \sigma(\eta_i) \end{cases} = \frac{e^{-\eta_i}}{1+e^{-\eta_i}} = \frac{1}{1+e^{+\eta_i}}$$

And we obtain the compact form:

$$p(y_i \mid x_i) = \sigma(y_i \eta_i) = \frac{1}{1 + e^{-y_i \eta_i}}$$

Realized average acceptance probability h

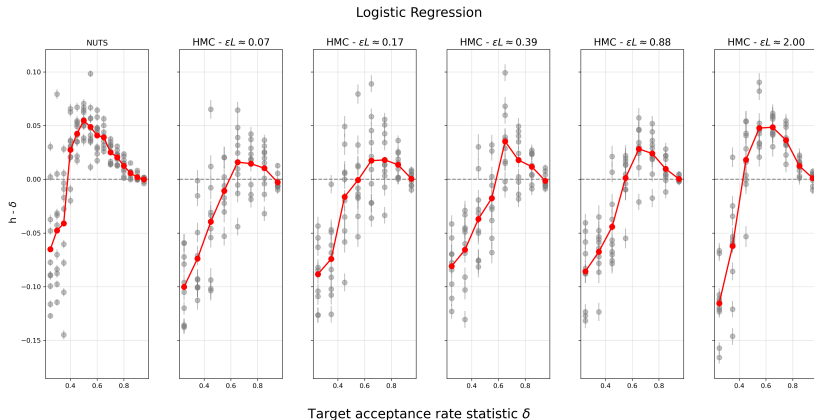
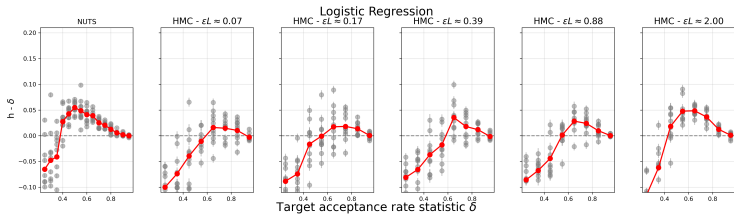
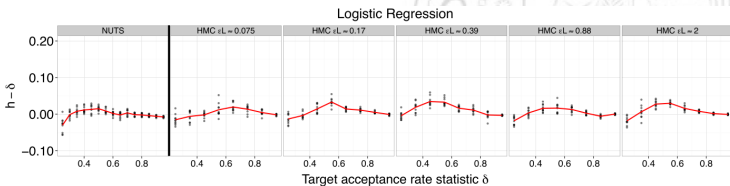


Figure: Each gray point is obtained by averaging across all 4 chains. The computation has been repeated 10 times for each value of δ . In red the average across all repetitions.

Acceptance probability comparison with paper



(a) Our results



(b) Paper's results.

ESS efficiency

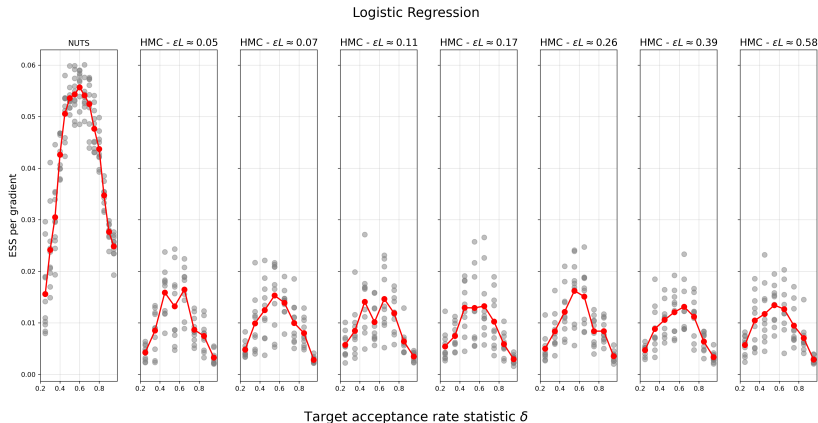
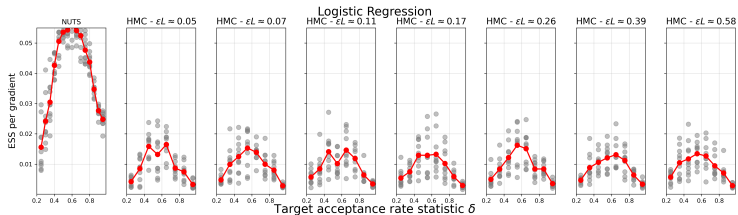
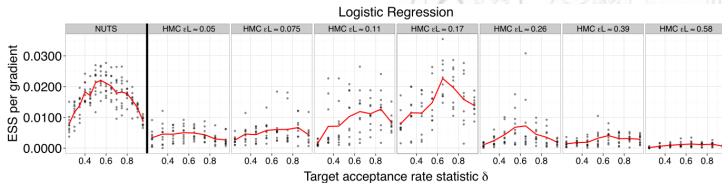


Figure: Each gray point is obtained by taking the minimum ESS per gradient across all 4 chains for all variables, considering both mean and variance. In red the average across all repetitions.

ESS efficiency comparison with paper



(a) Our results



(b) Paper's results.

Hierarchical Logistic Regression on German Credit Data



Changes from previous model

1. We expand the predictor vectors by including two-way interactions $(x_i \cdot x_j, i \neq j)$, resulting in $\binom{24}{2} = 276 + 24 = 300$ -dimensional vectors of predictors $\mathbf{x}$ and a 300-dimensional vector of coefficients β .
2. The variance parameter in the prior on α and β is given an exponential prior and estimated as well.
3. We set $\lambda = 0.01$, yielding a weak exponential prior distribution on σ^2 , whose mean and standard deviation are 100.



These changes make for a more challenging problem: the posterior is in higher dimensions and the variance term σ^2 interacts strongly with the remaining 301 variables.

The paper's posterior

$$p(\alpha, \beta, \sigma^2 \mid X, y) \propto p(y \mid X, \alpha, \beta) p(\beta \mid \sigma^2) p(\alpha \mid \sigma^2) p(\sigma^2) \propto \\ \exp \left(- \sum_i \log(1 + \exp\{-y_i x_i \cdot \beta\}) - \frac{\alpha^2}{2\sigma^2} - \frac{1}{2\sigma^2} \beta \cdot \beta - \frac{N}{2} \log \sigma^2 - \lambda \sigma^2 \right)$$

where $N = 1000$ is the number of customers.

However...

that should instead be $\frac{D+1}{2}$ where D is the dimension of β (300).

(Also the intercept term α is missing in the (log)-likelihood)

Motivation

Now the variance σ^2 on α and β is inferred from data, therefore in the log-posterior for these parameters we get:

$$\log p(\alpha \mid \sigma^2) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{\alpha^2}{2\sigma^2}$$

$$\log p(\beta \mid \sigma^2) = -\frac{D}{2} \log(2\pi\sigma^2) - \frac{\beta^T \beta}{2\sigma^2}$$

So their total contribution to the log-posterior is:

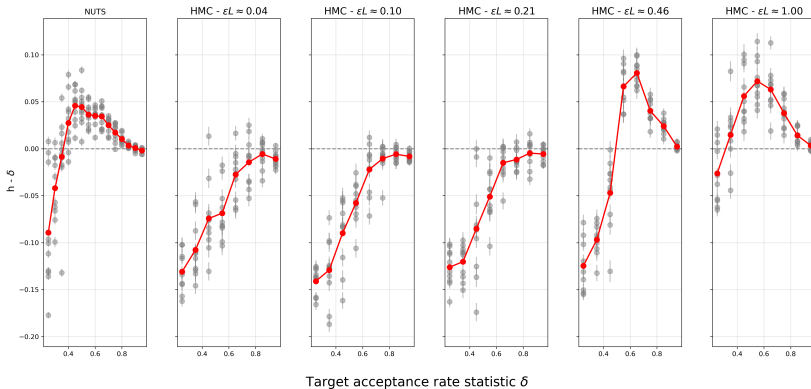
$$-\frac{\alpha^2 + \beta^T \cdot \beta}{2\sigma^2} - \frac{D+1}{2} \log(\sigma^2)$$

Therefore, the log-posterior we sampled from is proportional to:

$$-\sum_i \log(1 + \exp\{-y_i (\alpha + x_i \cdot \beta)\}) - \frac{\alpha^2}{2\sigma^2} - \frac{\beta^T \beta}{2\sigma^2} - \frac{D+1}{2} \log \sigma^2 - \lambda \sigma^2$$

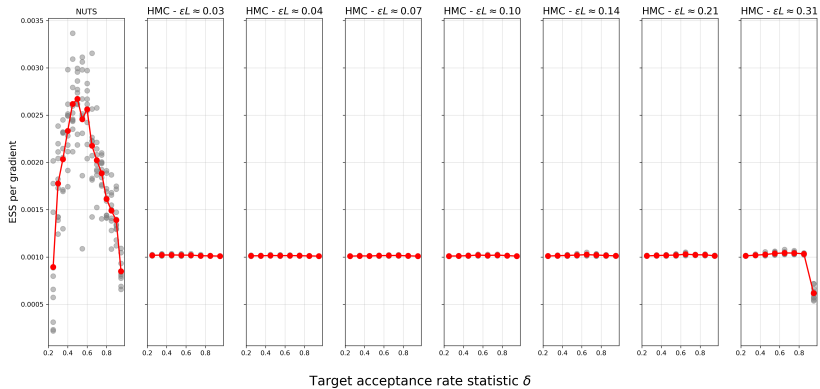
Realized average acceptance probability h

Hierarchical Logistic Regression

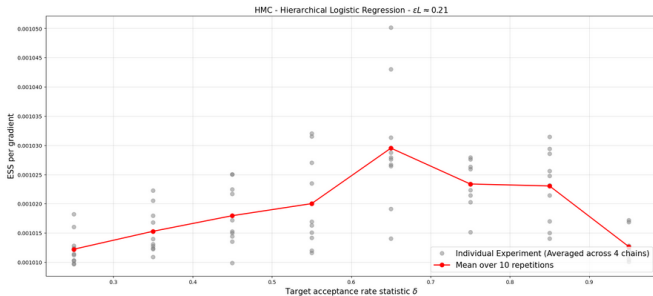
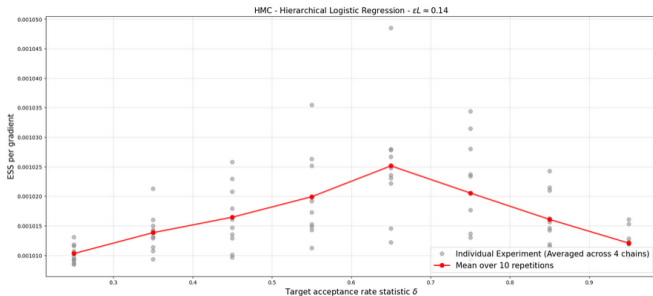


ESS efficiency

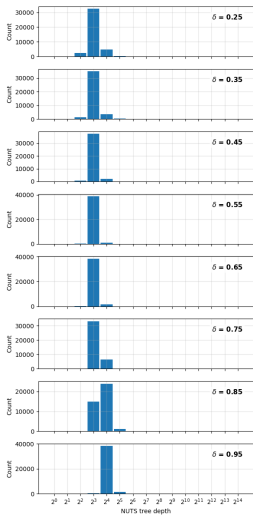
Hierarchical Logistic Regression



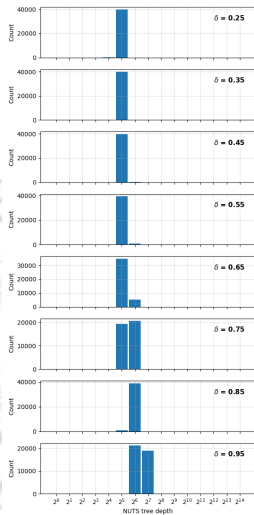
ESS efficiency: HMC example



NUTS Tree Depth Histogram



(a) Logistic Regression Tree Depths.



(b) Hierarchical LR Tree Depths.